

Handbook EGS3012S 2026: Atmospheric Science

Course convenor:



Dr. Stefaan Conradie (s.conradie@uct.ac.za); EGS building Rm 4.14

Lecturers:



Dr. Phikolomzi Matikinca (phikolomzi.matikinca@uct.ac.za; EGS 4.08)



Dr. Sabina Abba Omar (s.abbaomar@uct.ac.za; EGS 6.01)



Dr. Lynette van Schalkwyk (lynette.vanschalkwyk@uct.ac.za; EGS 6.01)

27 July 2026

If anything from here is not clear, please ask! We are here to support you.

Contents

1	Course Aim and Overview	4
1.1	Course Objectives	4
2	Practicals	5
2.1	Tutors	5
3	Assessment	6
3.1	Tests	6
3.2	Practicals	6
3.3	Project	7
3.4	DP - Duly performed requirements	7
3.5	Examination	7
3.6	Valid excuses for missed assessments	8
4	Calendar & Timetable	9
5	Amathuba	9
5.1	Amathuba Submissions	10
5.2	Late Submission	10
6	Referencing	10
7	Academic Integrity	11
7.1	Plagiarism	11
7.2	EGS AI statement and policy	11
7.2.1	Statement on the Responsible Use of Artificial Intelligence (AI)	11
7.2.2	Core principles	11
7.2.3	AI use in teaching and learning	12
7.2.4	Permitted low risk uses	12
7.2.5	Prohibited or high-risk uses	13
7.2.6	Declaration of AI use	13

7.2.7	AI use in research	13
7.2.8	Final position	14
7.3	Consequences of inappropriate AI use in EGS	14
7.4	Specific GenAI requirements for EGS3012S	14
8	Teaching Continuity	14
9	Suggested resources	15
10	Class Representative: TBC	15
10.1	Class WhatsApp group	15
10.2	Policy Document on Class Representatives (from SRC January 2020)	15
10.2.1	Introduction	15
10.2.2	Definition of a Class Representative	15
10.2.3	Aim	16
10.2.4	Roles and Responsibilities	16
10.2.5	Student Governance Support Structures to Class Representatives	16
10.2.6	Class representative System	17
10.2.7	Termination of Class Representatives	18
11	Core competencies in EGS	18
11.1	Questions to support critical assessment of transformation in EGS curriculum	19

List of Tables

1	EGS3012S 2026 final mark breakdown	6
2	Provisional course plan 2026	9

1 Course Aim and Overview

The overall aim of the course is to understand the **atmosphere** as a key and impactful element of the changing **climate system**, from a (southern) **African perspective**, by developing, verifying, interpreting, applying and extending **simple conceptual and mathematical models** of **atmospheric phenomena and their impacts**.

In the first two and a half weeks of the course, a multidisciplinary introduction to the science of climate variability and change is provided, with a focus on the resulting impacts on terrestrial, coastal, and marine environments. Students will explore both natural and anthropogenic drivers of climate change, assess key feedback mechanisms and impacts on ecosystems and human communities. The section integrates foundational scientific concepts with regionally and globally relevant case studies, encouraging students to critically engage with real-world environmental challenges. Emphasis is placed on both the biophysical processes and the socio- ecological dimensions of climate impacts.

By the end of the course you should be able to **explain** to a friend not in EGS3012, e.g.:

- What's the deal with the developing “**Super El Niño**”?
- Why the **climate system** is warming and how this affects **weather** and ecosystems.
- Why the mountain is often cloudy while the Cape Flats is sunny.
- Why the weather may seem pleasant in the morning at Tugwell, but then you arrive at South Stop and it's cold, cloudy, wet.
- Why Cape Town gets its rain in winter, Joburg and Durban in summer and Gqeberha throughout the year
- Why Cape Town's so windy & mainly from SE in summer; NW in winter
- Why thunderstorms happen mostly on summer afternoons
- Why some summers in **southern Africa** are wet with floods, while others are dry & hot
- Connect the dots between these questions

1.1 Course Objectives

To develop foundational understanding of interrelations between

1. What **atmospheric models** do
2. The **concepts of weather & climate** and the **relevant spatial & temporal scales**
3. The distinction between internal, natural climate **variability** and (human-driven) climate change.
4. The **impacts of climate change** on terrestrial and marine **ecosystems**
5. Perspectives regarding how to address **climate change**
6. Earth's **energy balance** & implications for its habitability

7. The atmosphere as a component of the (complex) **climate system**, the other subsystems of which it's constantly interacting with
8. Fundamental principles of **atmospheric thermodynamics**
9. Equations that describe **motion in the atmosphere** (climate and weather **dynamics**)
10. Reading **synoptic charts & maps** of other key weather variables, understanding how they relate to thermodynamic and dynamic concepts
11. **Weather forecast for South African cities** for 1–2 days

2 Practicals

Tuesday OR Wednesday 2-5pm weekly from 28 July In Studio 5

Prac day sign up (max 30) On Amathuba:

https://amathuba.uct.ac.za/d2l/lms/group/group_list.d2l?ou=208513&categoryId=11851

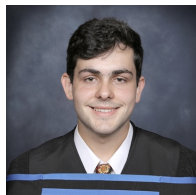
10 prac submissions Each counts 2% of course total: $2\% \times 10 = 20\%$

2.1 Tutors



Head tutor: Tanya van der Merwe (vmrtan005@myuct.ac.za)

Tutors: 3 Tuesday, 3 Wednesday To be confirmed for days



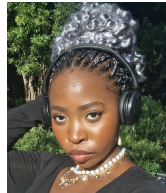
- Isaac Oosthuizen



- Megan van Schoor



- Affectionate Makamu
- Khanyisile Mpithi (no photo)



- Olebogeng Matlatla

Tutors will introduce themselves at the first prac.

3 Assessment

By the new Science Faculty rules, at least 80% of assessments must be invigilated in-person. That means at least 5 of the 10 pracs will be in-person, probably more. The project is a non-invigilated assessment.

task types	percentage of total
3 Tests	10%
Project	10%
10 Pracs	20%
Examination	60%

Table 1: EGS3012S 2026 final mark breakdown

3.1 Tests

Fridays, 8am-8:45am, Studio 5 (class time): 14 Aug, 25 Sep, 23 Oct

Please be ready by 7:55 latest. Arrive no later than 7:50

Will include MCQs, may include short answer questions

3.2 Practicals

DP requirement In-person attendance of all practical sessions (weekly) and submission of all prac activities

3.3 Project

Poster A0 size, virtual (not printed)

African extreme climate/weather event that occurred between 2016 and 2026

Individual submission on a unique event — ensure your topic is different to all other students

Released week 1 First prac will involve prep work for topic selection

Due 19 October 2026, 17:00 Serious attempt is a DP requirement

3.4 DP - Duly performed requirements

To be allowed to write the exam:

Attendance and completion of all class assignments (i.e. pracs, tests and project).

Class Record (CR) $\geq 40\%$

3.5 Examination

In-person, 3-hour Date, time and venue is decided by the UCT examination office and will be communicated in due course

Subminimum A student is required to achieve at least 40% in the examination

The exam must be written Applications for deferred exams are reviewed by a university examinations review panel, who are known to be strict and require extensive documentation to support applications

Supplementary and deferred exams The dates and times for supplementary (supp) and deferred exams (DEs) are set by the University, centrally, and are fixed and inflexible. Supplementary and deferred exams will be held in the UCT-scheduled timeslot, which will be between late November 2026 and mid-January 2027.

Grading for supp/deferred exams If you write a supp or DE, your grade will be calculated according to the same weighting outlined above in [Table 1](#), with the exam mark substituted with the supp/DE mark

Further details on Supplementary Exams For a 3rd year course, the Science faculty now allows the awarding of supps if a student has achieves at least 40% for their course mark and has achieved all other subminima (i.e. 40% for the exam). For example, this means if you get 48% overall, but if your exam mark is 38% this will result in an outright fail (48F for example), with no supp awarded. If a supp is passed, no quantitative grade is given only a generic “UP” grade (supplementary pass).

3.6 Valid excuses for missed assessments

Stefaan's interpretation of UCT policy on valid excuses is that there are five types of excuses that will be considered valid reasons to miss course assessments (note, for exams a university-wide policy applies). These are:

medical reason A doctor's medical/sick note is required, specifying that you were not well enough to attend the assessment in question and the time period during which you were unwell. This medical note must be uploaded on Amathuba, in the provided space under "Assignments", within 24hrs of the end of the assessment in question to be considered. The file name should include your student number and the assessment for which you are being excused, e.g. CNRWIL004_Test1_medicalnote.pdf. Medical certificates supporting late Submission need to cover the period leading up to the submission date in addition to the submission date itself.

compassionate reason If someone close to you has severely unwell or has passed away; appropriate documentation will also be required and should be submitted following the process above.

political or religious reasons If, for example, a holy day coincides with an assessment or you can provide clear and convincing reasons why you object on political grounds to being assessed at the time and place or manner specified. Such justifications need to be provided at least a week **in advance** of the assessment.

sporting reason If you are to represent South Africa or a provincial team competitively, this needs to be indicated through submitting appropriate documentation to the lecturer concerned and course convener at least a week **in advance** of the assessment.

technical or logistical reasons A thorough explanation of why it was not possible for you to submit, attend or do an assessed activity at the time or place required due to circumstances beyond your control should be submitted as detailed in the "medical reason" above, within 24hrs after the end of the assessment.

For consistency, concessions will not be considered for other excuses, in particular for concerts, festivals, general celebrations or ceremonies, etc.

If more than one test is missed with a valid reason, a student may be required to write a make-up test near the end of the fourth term.

Please note that even if valid reasons are provided in all cases, if sufficiently many assessments and classes are missed that the course management team and HoD agree that a student has very little chance of passing a course, they will not be granted a DP, to reduce pressure on the student during the exam period.

4 Calendar & Timetable

Lecture times: Monday - Friday, 8.00 – 08:45 am.

Lecture venue: EGS Building, Studio 5

Lecture format: Face-to-face

Lecture slides: Released on Amathuba at the end of the week

Additional notes will be available for some sections in a collated course reader to be released at the start of Stefaan's section

Table 2: Provisional course plan 2026

Week	Dates	Lecturer	Topic	assessments
1	27 - 31 Jul	PM	Climate Variability, Change & Impacts	Prac 1
			Also: class rep, project introduction	
2	3 - 7 Aug	PM	Climate Variability, Change & Impacts 2	Prac 2
3	11 - 14 Aug	PM & SC	Intro to atmospheric science, energy balance & models	Test1 - 14 Aug
			Monday 10 Aug is a public holiday	also: Prac 3
4	17 - 21 Aug	SC	Thermodynamics 1: temperature, pressure, density	Prac 4
			also: project topic selection deadline	
5	24 - 28 Aug	SC	Moist Thermodynamics	Prac 5
6	31 Aug - 4 Sep	SC	Lapse Rates & Stability; climate data; vectors, forces	Prac 6
	7 - 11 Sep		Mid-Semester Vacation	
7	14 - 18 Sep	SC	Dynamics: intro, geostrophic & gradient winds	Prac 7
8	21 - 25 Sep	SC	Kinematics, Global circulation	Test 2 - 25 Sep
			24th Thurs is public holiday	
9	28 Sep - 2 Oct	SC	Rossby wave dynamics & Mid-latitude weather systems	Prac 8
10	5 - 9 Oct	SC	Tropical & subtropical weather systems	Prac 9
11	12 - 16 Oct	LvS	SA weather & climate	Prac 10
12	19 - 23 Oct	SC & SAO	Climate data, Climate Modelling & Geo-engineering	Test 3 - 23 Oct
			Project submission deadline: 19 October	
13	26 - 30 Oct	SC	Predictability - 2 lectures, then consolidation	DP released
	2 - 3 Nov		Consolidation	Exam date: TBC

5 Amathuba

Take home assessments must be submitted via Amathuba in the appropriate assignment entry. Ask your lecturer, course convenor or tutor if you are unsure.

A general (but moderated, anonymous if you choose) Q&A is also available on the Amathuba site via the "Discussions" tab.

5.1 Amathuba Submissions

- Please do not e-mail electronic submission to staff e-mail accounts. Sorry, but we cannot submit on your behalf and to keep track of submissions and have submissions go through Turnitin, they have to be submitted on Amathuba only.
- Plagiarism will be tested for and where violations of academic integrity policies are identified or suspected, cases may be referred for formal disciplinary processes.
- All electronic submission must be in PDF (.pdf) format.
- Please ensure that submissions are no more than 5MB in size, wherever possible.
- Suggested file naming: STUDENTnum EGS3012S_Prac.n_YYYYMMDD.pdf, e.g. CNRWIL004_EGS3012S_Prac.3_20260827.pdf (NO SPACES!)

5.2 Late Submission

- We use Amathuba Time Stamp to assess late submission
- Late Submission will be subject to 5 % deduction per day for 4 days (96 hours)
- Thereafter, submissions will not be marked and receive a 0 grade.
- Resubmissions on Amathuba will be given a later time stamp and possible penalty.
- It is your responsibility to ensure that the submission is the correct, final version. Please do check!
- After marks are released, no new submissions will be considered.

6 Referencing

Individual lecturers may provide specific instructions for tasks they set, but in general, Stefaan as course convenor is not fussed about exactly which referencing system you use, but I do strictly require the following (you may be requested to resubmit an submitted work within 48hrs if you do not comply for the submission to be marked):

1. All statements that don't follow from evidence you present directly must be supported by an in-line citation, using either an author-date or numeric citation style
2. All works cited inline must appear in the bibliography
3. The style used across all citations and bibliography entries must be consistent
4. Bibliography entries that refer to resources for which an online abstract or similar is available (basically almost all sources, including journal articles, book chapters, websites, datasets, etc.) must include a DOI or URL for the resource, hyperlinked where possible.
5. Bibliography entries should include author details, year, title and information about where it was published, e.g. in a particular journal or book, or chapter.

7 Academic Integrity

Academic integrity involves, among other things, not presenting other people's work as your own — in other words not plagiarising.

7.1 Plagiarism

More information about plagiarism and how to avoid it can be found on Amathuba:

[Academic Integrity](#)

With all take-home or online submissions, you must include a plagiarism and AI use declaration, modified from the [template declaration available on Amathuba](#). Please read the document and modify as appropriate before signing; signing a document you have not read is a serious transgression.

For More Information visit the UCT Writing Centre 6th level Hlanganani Building, Upper Campus

A further particularly important component of academic integrity relates to the use of AI:



7.2 EGS AI statement and policy

7.2.1 Statement on the Responsible Use of Artificial Intelligence (AI)

Artificial intelligence (AI), including generative AI tools such as large language models, is now a routine part of academic, professional, and public life. The Department of Environmental & Geographical Science (EGS) recognises that AI tools can support aspects of learning, teaching, and research, while also posing significant risks to academic integrity, skill development, and ethical research practice.

This statement sets out the Department's position on the appropriate and responsible use of AI by students and staff in teaching, learning, and research. It should be read in conjunction with University of Cape Town policies, Faculty of Science guidance, and relevant ethics and integrity frameworks, including:

- [The University of Cape Town Senate Ethics in Research Committee \(EiRC\) Guidelines and Recommendations for the Use of Generative AI Tools in Research](#)
- Guidance from the Centre for Innovation in Learning and Teaching ([CILT](#)) on assessment and academic integrity in the context of AI use.

7.2.2 Core principles

The EGS Department approaches AI use through the following principles:

Human responsibility and accountability: AI tools do not replace scholarly judgement. Students and staff remain fully responsible for the accuracy, originality, and integrity of all submitted work.

Transparency: Any permitted use of AI must be openly declared in line with course, departmental, and university requirements.

Skill development: Core disciplinary skills, including critical reading, data analysis, writing, argumentation, field observation, and ethical reasoning, must be developed independently of AI.

Research integrity and ethics: AI use must align with UCT research ethics codes, authorship policies, and legal obligations, including data protection and consent requirements.

Equity and epistemic awareness: Users should be alert to bias, exclusion, and Global North dominance embedded in many AI systems, particularly in environmental and geographical research contexts.

7.2.3 AI use in teaching and learning

The Department shares the University's concern that inappropriate AI use can undermine student learning and lead to academic dishonesty. In line with Faculty of Science policy, the majority of summative assessment is conducted under controlled, in-person conditions.

The EGS Department's default position is that assessed work must reflect the student's own reading, analysis, interpretation, and writing, and must not be substantially generated by generative AI tools. This includes the generation of arguments, interpretations, or written text for submission.

Limited use of AI tools for specific purposes (for example, brainstorming broad topics, checking clarity or grammar, or exploring presentation formats) may be permitted where explicitly stated by the lecturer. Where such use is permitted, the scope and purpose of AI use will be clearly defined in the course or assignment instructions, and all use must be transparently declared.

The primary task of students in EGS remains the development of core academic skills independent of AI, including critical reading, data analysis, synthesis, and disciplinary writing.

7.2.4 Permitted low risk uses

Where explicitly authorised by the lecturer, AI tools may be used in limited, low-risk ways that support learning without substituting for core academic work. These may include:

- Brainstorming broad topics or research questions
- Improving clarity, grammar, or presentation of text written by the student
- Generating ideas for data visualisation or presentation formats
- Checking logic, calculations, or code, provided all outputs are independently verified

In all cases, students must ensure that they understand every step of the work and are able to explain and defend it without reliance on AI tools.

7.2.5 Prohibited or high-risk uses

Unless explicitly authorised in writing by the lecturer for a specific task, the following uses are not permitted in EGS courses:

- Generating full or partial assignments, essays, reports, or reflective pieces
- Producing arguments, interpretations, or conclusions for submission
- Summarising academic literature in a manner that bypasses learning or uncritically selecting readings for assessed work. Finding literature using appropriate/approved AI tools is acceptable.
- Analysing data or generating interpretations without human-led analysis
- Fabricating, manipulating, or obscuring sources, data, or methods
- Directly inputting the whole or sections of questions, exercise or task descriptions from UCT course assessments, as this violates UCT's IP policies
- Inputting confidential, sensitive, or personally identifiable data into public AI systems
- Such uses are likely to constitute academic misconduct or breaches of research ethics.

7.2.6 Declaration of AI use

Where AI use is explicitly permitted, students must include a clear declaration with their submission. This should state:

“I have used [name of AI tool], as permitted by the lecturer, for the purpose of [brief description of use]. I take full responsibility for the final content of this submission.”

Failure to declare AI use, or use of AI beyond what is permitted, will be treated in line with UCT's rules on plagiarism and academic misconduct.

7.2.7 AI use in research

For research activities, EGS aligns with UCT Senate Ethics in Research Committee guidance on generative AI. AI tools may support certain stages of the research process, but their use must be:

- Transparent and disclosed where relevant
- Ethically reviewed where human, animal, or sensitive data are involved
- Consistent with authorship, accountability, and data protection requirements

Researchers remain fully accountable for AI-assisted outputs, including errors, bias, or misrepresentation.

7.2.8 Final position

The EGS Department recognises the value of AI literacy as a graduate attribute, but prioritises deep disciplinary understanding, ethical practice, and intellectual independence. AI tools may support learning and research when used carefully and transparently, but they must never replace critical thinking, scholarly responsibility, or human judgement.

Students and staff are encouraged to discuss uncertainties about AI use with lecturers or supervisors before proceeding.

7.3 Consequences of inappropriate AI use in EGS

The EGS Department’s position on the consequences of inappropriate use of AI is guided by [the UCT Policy for the Prevention and Management of Academic Misconduct by Students](#). The Department will evaluate the severity of the violation and act accordingly.

According to the UCT policy, students “will only be given credit for work that is their own and work that has been duly and appropriately referenced. . . . [O]nly those portions of the work that are the student’s own work and have been appropriately referenced should be awarded any marks. Portions that are not identifiable as the student’s own work and are not duly referenced should not be awarded any marks. Where it is not possible to determine which portions are the student’s own work with reasonable ease and certainty, or none of the work is the student’s own, it will not be possible to award a mark to the work submitted. The only possibility in such cases is to give it a mark of zero.”

Additionally, the Department may make a decision to refer the student for formal disciplinary action.

7.4 Specific GenAI requirements for EGS3012S

If you choose to use GenAI tools for a submission, which is allowed for take-home submissions subject to the conditions outlined in [subsection 7.2](#) above, you are required to fulfil an additional essential requirement: you endeavour to note all uses of GenAI tools, *record conversations verbatim* and submit them as *appendices* to submissions. Failing to do so will result in you being contacted to resubmit within 48hrs of the submission deadline, failing which your submission will not be marked.

8 Teaching Continuity

As per Science Faculty policy, if lectures are disrupted and/or access to campus is restricted, where possible, teaching and learning will continue online, with live online lectures via Microsoft Teams. Online lectures will be recorded as well.

The course convenor will make announcements about changes to the mode or timing of any activities associated with the course. Students without Internet access at home should take note of [eduroam locations](#) and other places where free WiFi is available.

9 Suggested resources

Freely available:

- [Roland Stull, 2017. *Practical Meteorology*](#)
- meted.ucar.edu (Free account sign-up required; lots of valuable resources, questions, etc.)

Available in UCT library:

- [Tyson & Preston-Whyte, 2000/1988. *The weather and climate of southern Africa*](#)
- [Van Zyl, 2003. *South African weather and atmospheric phenomena*](#).
- [Peixoto & Oort, 1992. *Physics of climate*](#) (! more advanced !)
- [McIlveen, 1992. *Fundamentals of weather and climate*](#)
- [Hobbs & Wallace, 2006. *Atmospheric Science: An Introductory Survey*](#)
- [Hidore et al., 2010. *Climatology: an atmospheric science*](#).

10 Class Representative: TBC

10.1 Class WhatsApp group

The class rep will be selected in the first week of class. They may set up a WhatsApp group which students in the course will be invited to join (not the lecturers or tutors), but joining this group is voluntary. Members of the group agree to comply with university policies around appropriate engagement with other group members and agree that disciplinary actions may be taken if they are reported anonymously for not doing so.

10.2 Policy Document on Class Representatives (from SRC January 2020)

10.2.1 Introduction

Class Representatives are democratically elected students dedicated to ensuring that the learning process and environment of their constituencies achieve the required level of excellence.

10.2.2 Definition of a Class Representative

A student enrolled in a course, democratically elected by the class, to represent the class in relation to course matters and to facilitate effective communication between students and staff as well as between students and the student governance structures.

10.2.3 Aim

1. To allow for greater engagement between staff and students so as to ensure an outstanding learning environment.
2. To ensure a continued flow of information with the student governance structures.
3. To empower students with regard to their academic experiences and careers.
4. To identify and address the concerns of students and convey complaints, issues and proposals along the correct channels.

10.2.4 Roles and Responsibilities

1. To be available to the class and determine what issues and concerns the class may have.
2. To convey the views of the class to course convenors, lecturers and tutors.
3. To direct issues or concerns that are unable to be resolved or which are still persisting to the Science Students' Council or Student Representative Council (SRC).
4. To inform students of any important news or events that the Student Representative Council or Science Students' Council need students to be aware of.
5. To regularly report back to the Science Students' Councils with regard to any challenges or recommendations.
6. To attend class representative workshops and faculty meetings arranged by the Science Students' Council and the Students' Representative Council.
7. *[EGS specific: to attend the September EGS staff meeting and report back on student general student questions, experiences & concerns in each course]*
8. To provide feedback to the class represented on the outcome of meetings with lecturers or the faculty.

10.2.5 Student Governance Support Structures to Class Representatives

Academic Representative of the Science Students Council Elected Student Representative with the portfolio of Academic Representative at Faculty level, who ensures the election of Class Representatives, ensures the proper training of Class Representatives and finally formally supervises the operation of the Class Representatives.

Science Students Council The Science Students Council consist of 10 democratically elected students dedicated to ensuring that student's interests are put forward at Faculty level. The Council also aims to ensure that class representatives are appropriately inducted and is responsible for reviewing the class representative system and making the necessary amendments and recommendations for improvement.

Undergraduate Academic Sub-Council of the Student Representative Council The Undergraduate Academic Sub-Council of the SRC consists of the Chairpersons of the 6 Undergraduate Faculty Councils as well as the Undergraduate Academic Chairperson of the Student Representative Council.

The Undergraduate Academic Sub-Council of the Student Representative Council strives to ensure that the University achieves its mission by ensuring that there is adequate representation on important Academic Committees both on a Faculty and University- wide level, ensuring that Undergraduate Faculty Councils are functioning effectively and providing support where it is necessary. The Council is also responsible for increasing awareness of various issues of academic importance and to ensure that student issues that arise are sent (via the Chairperson) to the relevant committee or person.

Chair of Undergraduate Academics of the Student Representative Council Elected Student Representative with the portfolio of Academic Chair at University level, who chairs the Undergraduate Academic Sub-Council of the SRC and serves as a direct link to the Student Representative Council executive and University Senate.

Student Representative Council The SRC consists of 17 democratically elected students dedicated to address all spheres of student interests at a University-wide level.

The SRC is the highest decision-making student body and has seats on important University committees; they can thus be referred to as the link between the students and University management.

Department of Student Affairs The Department of Student Affairs will assist the above-mentioned Student Governance Structures with administrative and supportive tasks, especially with regard to the training of the Class Representatives.

10.2.6 Class representative System

Election Elections are to be run by the course convenor who will request that the class either nominate individuals for Class Representative or individuals volunteer themselves. Students standing for election are then to motivate their candidacy and the election may take place by show of hands or by secret ballot, whichever the class deems more appropriate. The elected students contact information must then be distributed to the class as well as the Academic Representative of the Science Students Council.

Training The Academic Representative of the Science Students Council with the assistance of the Science Students Council, the Chair of Undergraduate Academics of the SRC and the Department of Student Affairs will provide the elected Students with training to allow for effective facilitation of their representative responsibilities.

Election of Departmental Representatives At the training of the Class Representatives the Academic Representative of the Science Students Council will facilitate the election of a Departmental Representative for each Department within the Faculty.

The Departmental Representative will serve as the direct link between the Academic Representative of the Science Students Council and the Class Representatives. It is the role of the Departmental Representative to assist the Class Representative in their representative endeavours as well as to supervise the Class Representatives in their Department so as to ensure effective student governance and representation of student academic concerns.

The Complaint Channels As follows:

1. The Class Representative shall first and foremost address the Lecturer and/or Course Convenor with regard to an issue, complaint or proposal that the class may have.
2. Should the issue remain unresolved the Departmental Representative, with the assistance of Class Representative, shall approach the Head of Department.

3. Should the issue continue to persist the Class Representative and Departmental Representative shall approach the Academic Representative of the Science Students Council and request that the said representative approach the Head of Department, and if still unresolved the Dean.
4. Upon continued persistence of the unresolved issue the Academic Representative of the Science Students Council with the Chair of the Science Students Council shall approach the Chair of Undergraduate Academics of the Student Representative Council, who will approach the Dean and then the executive management of the University should the issue continue to remain unresolved.

Meetings The Departmental Representatives should make sure to touch base with their Departments Class Representatives on a regular basis. Further, the Departmental Representatives should regularly meet with Academic Representative of the Science Students Council to address trending concerns in their departments.

10.2.7 Termination of Class Representatives

A Class Representative ceases to hold their position if or when:

1. the course for which they have been elected Class Representative concludes
2. they cease to be a student at the University of Cape Town;
3. they tender their resignation in writing and the Science Students Council accepts the resignation. Following this, the Class Representative is obligated to find a replacement within their class to represent their class;
4. they have had two motions of censure passed on them during the course of the year pertaining to:
 - 2.4.1 failure to attend two or more consecutive meetings without an acceptable apology in advance
 - 2.4.2 failure to carry out duties entrusted to them by the Science Students Council
 - 2.4.3 bringing the name of the Science Students Council into disrepute by being found guilty of any serious offence under the University's disciplinary system or South African law

11 Core competencies in EGS

A graduate with an EGS major should (as determined by the department generally, not the EGS3012S team specifically):

1. Have a holistic view of the environment and its physical, biological, social, economic and political components in time and space.
2. Have an awareness of the history and philosophy of geography.
3. Recognise that knowledge and understanding are dynamic entities.
4. Understand and interpret the dynamic patterns and processes underlying landscapes at various scales.

5. Possess an awareness of the spatial manifestations, dimensions and implications of physical, social, economic, political and cultural processes.
6. Be competent in the methodology of research and be able to conduct critical enquiries pertaining to environmental and geographical phenomena and concepts.
7. Be able to access appropriate information from a variety of sources including libraries, archives, public information data bases and the internet.
8. Have the skill to acquire, organise, analyse and synthesise information in order to solve environmental and geographical problems.
9. Have knowledge of key geographical, biophysical and social theories and concepts and be able to apply such understanding to practical problems in geography.
10. Understand the concept of sampling and be able to apply descriptive and analytical statistics to geographical data.
11. Be able to understand and use maps, diagrams and other forms of representation of geographical data.
12. Take measurements from maps and remote sensing products, develop basic cartographic representations and use Geographical Information Systems to represent and analyse spatial data.
13. Interpret maps and remote sensing products and make inferences from these representations.
14. Be proficient with tools and techniques of qualitative analysis (e.g. interviews and questionnaires), be able to classify, analyse, and summarise data and draw inferences from the results.
15. Make basic field and laboratory observations and measurements, represent these in the form of field and laboratory notes, sketches, and interpret and analyse the results.
16. Work effectively as a member of a team or group in projects and investigations.
17. Demonstrate effective information and communication technology skills.
18. Communicate geographical understanding in writing, orally and using visual, symbolic and/or other forms of representation.
19. Exhibit proficiency in self-management, interpersonal skills and written and oral communication.
20. Manage and organise learning activities responsibly taking into account due ethical and cultural considerations.

11.1 Questions to support critical assessment of transformation in EGS curriculum

Curriculum content Questions:

- 1.1 How does our curriculum critically reflect on and respond to the history of the disciplines of Geography and Environmental Science in terms of dominant paradigms and approaches?
- 1.2 Does the curriculum adequately reflect our location in Africa and the global South?
- 1.3 To what extent is scholarship from Africa and the global south embedded in the curriculum?

- 1.4 How does our curriculum regard Africa and the South as sites of theoretical production and not just application—as sites to learn from and not just learn about?
- 1.5 Are there gaps in the curriculum in terms of theories taught and topics covered that would be especially important given our context in global south?

Pedagogical practices and learning methodologies Questions:

- 2.1 We recognise that our learning spaces inherit the social hierarchies (e.g. first language English speakers interact with certain lecturers more easily than 2nd/3rd/+ language English speakers) and divisions that exist in our society
- 2.2 What strategies and pedagogic practices can we put in place to deal with these divisions and challenges in the classroom?
- 2.3 What pedagogic strategies can we use to draw on the lived experiences that different students bring to the classroom?

Student support with the curriculum Questions:

- 3.1 What kinds of support do students need in order to engage fully with our courses (e.g. with literature, assignments etc.)
- 3.2 What can the department do to encourage retention of ENGEO students at undergraduate level and embark on postgraduate studies in the department?
- 3.3 What actions are needed to improve communication and relationships between students and lecturers?

General Questions:

- 4.1 To what extent does your worldview and context shape the way you structure and teach your course?
- 4.2 What pedagogic strategies can we use to draw on the lived experiences that different students bring to the classroom?
- 4.3 What can the department do to facilitate feelings of equality amongst all students?